SOEN 342

Project Iteration 1

Date: Friday, October 23rd

Name	Student ID
Adam Benchekroun	40306874
Mountaga Sy	40312584
Abderraouf El Khalil Karoun	40315753

Use Cases

#	Use case	Class	Relationship	Basis in the text
1	Register patient	Non-critical	Independent	Patients are uniquely identified, system maintains patient information
2	Search patient	Non-critical	Independent	Search by insurance number, date of birth, phone, partial name
3	Book appointment	**Critical**	Includes 4 and 5	Patients or authorized staff request appointments
4	Check availability and constraints	**Critical**	Included in 3	System verifies availability and scheduling constraints
5	Verify referral requirement	**Critical**	Included in 3	Appointment may require a referral depending on the service
6	Confirm appointment	Non-critical	Independent	Appointments may be confirmed
7	Reschedule appointment	Non-critical	Independent	…rescheduled
8	Cancel appointment	Non-critical	Independent	…or cancelled
9	Register patient arrival	**Critical**	Independent	Administrative staff register arrival
10	Conduct consultation and record clinical info	**Critical**	Extended by 11 and 12	Observations, diagnoses, treatment recommendations, follow-up
11	Issue prescription	**Critical**	Extends 10	Professional may issue a prescription
12	Request follow-up appointment	Non-critical	Extends 10	…or request a follow-up appointment
13	Issue referral	**Critical**	Independent	Referral records referrer, specialty, reason, priority, status
14	Update referral status	Non-critical	Independent	Referral has a status; may result in appointments
15	Generate invoice	**Critical**	Independent	Record fees, generate an invoice
16	Register payment	**Critical**	Independent	…and register payments
17	Manage professionals (info and schedule)	Non-critical	Independent	Professionals have information, specialties, schedules, locations

18	Manage user roles and access	Non-critical	Independent	Access depends on the user's role
19	View activity log	Non-critical	Independent	Activities recorded for accountability

Use Case Diagram

https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/Use_Cases_Diagram.png

Domain Model

https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/domain_model.png

Sequence Diagrams for Critical Use Cases

UC #	Use case	Class	Link to SSD
3	Book appointment	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/sequence_diagram_book_appointment.png
4	Check availability and constraints	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/UC4%20Check%20Availability.png
5	Verify referral requirement	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/UC5%20Verify%20Referral.png
9	Register patient arrival	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/sequence_diagram_register_patient.png
10	Conduct sconsultation and record clinical info	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/UC10%20Conduct%20Consultation.png
11	Issue prescription	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/Issue_Prescription_SSD.png
13	Issue referral	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/sequence_diagram_issue_referral.png
15	Generate invoice	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/Generate_Invoice_SSD.png
16	Register payment	**Critical**	https://github.com/Adambench/Soen-342-Project/blob/main/artifacts/Register_Payment_SSD.png

Fully-Dressed Scenarios (UC4, UC5 and UC10)

Use Case UC4: Check Availability and Constraints

Scope: Appointment Tracking System

Level: Subfunction (included in UC3 Book Appointment)

Primary Actor: Requester (patient or administrative staff)

Stakeholders and Interests:

- Patient: wants an appointment with the right professional at a time that suits them, without being double-booked.
- Administrative staff: wants a quick answer, and alternatives when the requested slot is taken.
- Healthcare professional: wants appointments only within working hours, at locations where they practise, for services they offer.
- Clinic: wants schedules filled without conflicts.

Preconditions: UC3 Book Appointment is in progress and has started an appointment with status “Requested”. The requester is logged in. The patient is identified. Professionals’ schedules, services and locations are recorded.

Success Guarantee: An available slot (professional, service, location, date, time) that meets every scheduling constraint is recorded on the appointment under way.

Main Success Scenario:

1. The requester enters the healthcare professional, medical service, clinic location, date and time for the identified patient.
2. The system verifies that the professional practises at that clinic location and offers that medical service.
3. The system verifies that the requested time is within the professional’s working hours and that the slot is not already booked.
4. The system verifies that the patient has no other appointment at that time.

5. The system confirms the slot is available, records it on the appointment and shows the professional, location, date and time.
6. The use case ends and control returns to UC3 Book Appointment.

Extensions:

2a. The professional does not practise at this location or does not offer this service:

1. The system reports that the service is not offered by this professional at this location.
2. The use case ends in failure and nothing is saved.

3a. The professional is unavailable (outside working hours or slot already booked):

1. The system reports that the slot is unavailable and lists the next available slots that meet every constraint.
2. The requester picks one of the proposed slots.
3. The use case resumes at step 5.

3a.2a. The requester picks none of the proposed slots:

1. The use case ends in failure and nothing is saved.

4a. The patient already has an appointment at that time:

1. The system reports a scheduling conflict for the patient.
2. The use case ends in failure and nothing is saved.

Special Requirements:

- Proposed slots are shown as a numbered list in the command-line interface, earliest first.

Technology and Data Variations List:

- Patient, professional, service and location are passed by their system IDs.
- Dates are entered as YYYY-MM-DD and times as HH:MM.

Frequency of Occurrence: Every time an appointment is booked.

Open Issues:

- Does each medical service have its own appointment length?
- How many alternative slots should the system propose, and how far ahead should it search?

Use Case UC5: Verify Referral Requirement

Scope: Appointment Tracking System

Level: Subfunction (included in UC3 Book Appointment)

Primary Actor: Requester (patient or administrative staff)

Stakeholders and Interests:

- Patient: wants to be booked without delay, with an existing referral recognized.
- Administrative staff: wants a clear answer on whether the booking can go ahead.
- Receiving professional: wants referral-only services booked only with a valid referral.
- Referring professional (clinic or external physician): wants the referral to lead to an appointment.
- Clinic: wants referral rules applied the same way every time.

Preconditions: UC3 Book Appointment is in progress. The requester is logged in. The patient is identified and an available slot has been selected (UC4).

Success Guarantee: Either the service requires no referral, or a valid open referral for the patient is linked to the appointment under way.

Main Success Scenario:

1. The requester asks the system to verify the referral requirement for the patient, the chosen service and the professional.
2. The system determines that the medical service requires a referral.
3. The system finds the patient's open referrals that request this professional's specialty or this professional.
4. The system displays the matching referrals: referring professional, requested specialty, clinical reason, priority and status.

5. The requester selects the referral to use.
6. The system links the referral to the appointment and confirms the link.
7. The use case ends and control returns to UC3 Book Appointment.

Extensions:

2a. The service does not require a referral:

1. The system reports that no referral is required.
2. The use case ends and the booking proceeds.

3a. The patient has no referral on file:

1. The system reports that a referral is required and the appointment cannot be booked.
2. The use case ends in failure and nothing is saved.

3b. The patient's referrals on file are not valid for this service (wrong specialty, cancelled or closed):

1. The system reports that the referral is not valid for this service.
2. The use case ends in failure and nothing is saved.

Special Requirements:

- Referral details shown to administrative staff follow the role-based access rules for clinical information.

Technology and Data Variations List:

- The referring professional may be a clinic professional or an external physician.
- A referral may request a specialty or a named professional.

Frequency of Occurrence: Every time an appointment is booked. A referral is needed only for services marked as requiring one.

Open Issues:

- Should administrative staff see the referral's clinical reason?
- Does a referral expire after a set time?

Use Case UC10: Conduct Consultation and Record Clinical Info

Scope: Appointment Tracking System

Level: User goal

Primary Actor: Healthcare professional

Stakeholders and Interests:

- Healthcare professional: wants the patient's history on hand and a quick way to record findings.
- Patient: wants an accurate record of diagnosis and treatment, seen only by authorized people.
- Administrative staff: wants the appointment marked Completed so the schedule and any invoice are accurate.
- Clinic: wants an accountable record of who accessed or changed clinical information.

Preconditions: The healthcare professional is logged in. The appointment exists and is assigned to them.

Success Guarantee: A consultation with the observations, diagnoses, treatment recommendations and follow-up requirements is saved to the patient's clinical record. The appointment status is Completed. Each access and change is recorded in the activity log.

Main Success Scenario:

1. The healthcare professional selects the patient's appointment and starts the consultation.
2. The system verifies that the user's role gives access to clinical information and that the patient's arrival has been registered.
3. The system displays the patient's information and clinical record history, and records the access in the activity log.
4. The professional enters the observations, diagnoses, treatment recommendations and follow-up requirements.
5. The system records the clinical information and confirms.
6. The professional ends the consultation.
7. The system saves the consultation to the patient's clinical record, sets the appointment status to Completed and records the changes in the activity log.

Extensions:

2a. The user's role does not give access to clinical information:

1. The system denies access and records the attempt in the activity log.
2. The use case ends in failure and nothing is saved.

2b. The patient's arrival has not been registered, or the appointment was cancelled:

1. The system reports that the consultation cannot start because the patient is not checked in.
2. The use case ends in failure and nothing is saved.

5a. Medication is needed:

1. The professional issues a prescription (UC11 Issue Prescription).
2. The use case continues at step 6.

5b. A follow-up is needed:

1. The professional requests a follow-up appointment (UC12 Request Follow-up Appointment).
2. The use case continues at step 6.

Special Requirements:

- Only roles authorized for clinical information can open a consultation.
- Every access to and change of clinical information is logged with date, time, user and description.

Technology and Data Variations List:

- Clinical fields are entered as free text.

Frequency of Occurrence: Once per attended appointment.

Open Issues:

- Can clinical information be corrected after the consultation ends, and by whom?
- If the professional exits without ending the consultation, is the entered information kept as a draft?

System Operations and Operation Contracts

System Operation

1. registerPatient(firstName, lastName, dateOfBirth, phoneNumber, insuranceNumber)
2. searchPatient(criteria)
3. makeNewAppointment()
4. checkAvailability(patientId, professionalId, serviceId, locationId, date, time)
5. selectSlot(date, time)
6. verifyReferralRequirement(patientId, serviceId, professionalId)
7. selectReferral(referralId)
8. endBooking()
9. confirmAppointment(appointmentId)
10. rescheduleAppointment(appointmentId, date, time)
11. cancelAppointment(appointmentId)
12. registerArrival(appointmentId)
13. startConsultation(appointmentId)
14. recordClinicalInfo(observations, diagnoses, treatmentRecommendations, followUpRequirements)
15. endConsultation()
16. startPrescription(consultationId)
17. addMedication(medicationId, dosage, frequency, duration)
18. finalizePrescription()
19. requestFollowUp(date, time)
20. issueReferral(patientId, issuerId, specialtyId, clinicalReason, priority)
21. updateReferralStatus(referralId, status)
22. startInvoice(appointmentId)
23. recordFee(description, amount)
24. generateInvoice()

25. selectInvoice(invoiceNumber)
26. registerPayment(amount, paymentMethod)
27. addProfessional(name, professionalInfo, specialtyIds, locationIds, workingHours)
28. updateProfessional(professionalId, name, professionalInfo, specialtyIds, locationIds, workingHours)
29. assignRole(userId, roleName)
30. viewActivityLog(fromDate, toDate)

Operation Contracts

Conventions (these apply to every contract below and are not repeated)

- The user is logged in, and their Role allows the operation.
- Every operation that changes state also creates an Activity (date, time, description). That Activity is associated with the user who performed the operation (Performs).
- Abbreviations:
    - p Patient, a Appointment, hp HealthcareProfessional, s AdministrativeStaff
    - ms MedicalService, loc ClinicLocation, r Referral
    - c Consultation, cr ClinicalRecord, pr Prescription, m Medication
    - i Invoice, pay Payment
- Appointment status is one of Requested, Booked, Confirmed, Arrived, Completed or Cancelled.
- Referral status is one of Open, Closed or Cancelled.

CO1 registerPatient(firstName, lastName, dateOfBirth, phoneNumber, insuranceNumber)
Cross references: UC1 Register patient
Preconditions: No patient with the same insuranceNumber exists.
Postconditions:

- Patient p was created with a new unique patientId. Its other attributes were set from the parameters.
- ClinicalRecord cr was created and associated with p (Has).

CO2 searchPatient(criteria)
Cross references: UC2 Search patient
Preconditions: The criteria are one of the following: an insurance number, a date of birth, a

phone number, or a partial first and/or last name.
Postconditions: None (query). The system returns the matching patients.

CO3 makeNewAppointment()
Cross references: UC3 Book appointment
Preconditions: None.
Postconditions:

- Appointment a was created with a.status = “Requested”.
- If staff are booking, a was associated with s (Manages).

CO4 checkAvailability(patientId, professionalId, serviceId, locationId, date, time)
Cross references: UC4 Check availability and constraints (included in UC3)
Preconditions: a is under way (CO3). p, hp, ms and loc exist.
Postconditions:

- These changes happen only if all of the following hold:
  - hp practises at loc and offers ms;
  - the time falls within hp’s Schedule.workingHours;
  - neither hp nor p has another appointment at that time.
- When they hold:
  - a was associated with p (Books), hp (Conducts), ms (Is-for) and loc (Held-at);
  - a.date and a.time were set.
- Otherwise nothing changed. The system returns the error. If only the slot is taken, it also returns the next available slots.

CO5 selectSlot(date, time)
Cross references: UC4 Check availability and constraints
Preconditions: CO4 reported the slot as taken and proposed alternatives. The chosen date and time are one of those alternatives.
Postconditions: The same as a successful CO4, using the chosen date and time.

CO6 verifyReferralRequirement(patientId, serviceId, professionalId)
Cross references: UC5 Verify referral requirement (included in UC3)
Preconditions: a has a slot (CO4 or CO5).
Postconditions: None (query). The system returns one of three results:

- “No referral required”, if ms.requiresReferral is false.
- Otherwise, the open referrals that concern p and request hp’s specialty.
- An error, if there are no such referrals.

CO7 selectReferral(referralId)
Cross references: UC5 Verify referral requirement
Preconditions: ms.requiresReferral is true. r is one of the referrals returned by CO6.
Postconditions: r was associated with a (Results-in). Nothing else about r changed.

CO8 endBooking()
Cross references: UC3 Book appointment
Preconditions: a has a slot (CO4 or CO5). If ms requires a referral, a has a linked referral (CO7).
Postconditions: a.status became “Booked”.

CO9 confirmAppointment(appointmentId)
Cross references: UC6 Confirm appointment
Preconditions: a exists and a.status = “Booked”.
Postconditions: a.status became “Confirmed”.

CO10 rescheduleAppointment(appointmentId, date, time)
Cross references: UC7 Reschedule appointment
Preconditions: a.status is “Booked” or “Confirmed”. The new date and time meet the same constraints as CO4.
Postconditions: a.date and a.time took the new values. a.status became “Booked”.

CO11 cancelAppointment(appointmentId)
Cross references: UC8 Cancel appointment
Preconditions: a.status is “Requested”, “Booked” or “Confirmed”.
Postconditions: a.status became “Cancelled”.

CO12 registerArrival(appointmentId)
Cross references: UC9 Register patient arrival
Preconditions: a.status is “Booked” or “Confirmed”, and a.date is today.
Postconditions: a.arrivalTime became the current time. a.status became “Arrived”.

CO13 startConsultation(appointmentId)
Cross references: UC10 Conduct consultation and record clinical info
Preconditions: hp conducts a.
Postconditions:

- If hp’s Role gives clinical access and a.status = “Arrived”, Consultation c was created and associated with a (Results-in).

- If hp's Role has no clinical access, the only change is an Activity recording the denied attempt.
- If a.status is not "Arrived", nothing changed.

CO14 recordClinicalInfo(observations, diagnoses, treatmentRecommendations, followUpRequirements)
Cross references: UC10 Conduct consultation and record clinical info
Preconditions: c is under way (CO13).
Postconditions: c.observations, c.diagnoses, c.treatmentRecommendations and c.followUpRequirements took the given values.

CO15 endConsultation()
Cross references: UC10 Conduct consultation and record clinical info
Preconditions: c is under way and its clinical information is recorded (CO14).
Postconditions:

- c was associated with cr, the ClinicalRecord of p (Contains).
- a.status became "Completed".

CO16 startPrescription(consultationId)
Cross references: UC11 Issue prescription (extends UC10)
Preconditions: c is under way. hp's Role allows prescribing.
Postconditions: Prescription pr was created and associated with c (Issues).

CO17 addMedication(medicationId, dosage, frequency, duration)
Cross references: UC11 Issue prescription
Preconditions: pr is under way. Medication m exists.
Postconditions: m was associated with pr (Lists), with the given dosage, frequency and duration.

CO18 finalizePrescription()
Cross references: UC11 Issue prescription
Preconditions: pr lists at least one Medication.
Postconditions: pr.date became the current date.

CO19 requestFollowUp(date, time)
Cross references: UC12 Request follow-up appointment (extends UC10)
Preconditions: c is under way for a. The date and time meet the same constraints as CO4.
Postconditions:

- Appointment f was created with f.date and f.time set and f.status = “Booked”.
- f was associated with c (Requests-follow-up).
- f was associated with a’s p (Books), hp (Conducts), ms (Is-for) and loc (Held-at).

CO20 issueReferral(patientId, issuerId, specialtyId, clinicalReason, priority)
Cross references: UC13 Issue referral
Preconditions: p and the Specialty exist. The issuer is a HealthcareProfessional or an ExternalPhysician already on file.
Postconditions:

- Referral r was created with r.clinicalReason and r.priority set and r.status = “Open”.
- r was associated with p (Concerns), with the issuer (Issues) and with the Specialty (Requests).

CO21 updateReferralStatus(referralId, status)
Cross references: UC14 Update referral status
Preconditions: r exists. The new status is “Open”, “Closed” or “Cancelled”.
Postconditions: r.status became the new status.

CO22 startInvoice(appointmentId)
Cross references: UC15 Generate invoice
Preconditions: a.status = “Completed”. ms is uninsured for p (not covered by RAMQ). a has no Invoice yet.
Postconditions: Invoice i was created and associated with a (Billed-by). i.fees is empty and i.total = 0.

CO23 recordFee(description, amount)
Cross references: UC15 Generate invoice
Preconditions: i is under way and not yet generated. amount > 0.
Postconditions: The fee was added to i.fees. i.total increased by amount.

CO24 generateInvoice()
Cross references: UC15 Generate invoice
Preconditions: i.fees holds at least one fee.
Postconditions: i.date became the current date. i received a unique invoice number.

CO25 selectInvoice(invoiceNumber)
Cross references: UC16 Register payment
Preconditions: i exists. Its balance (i.total minus its payments) is above 0.
Postconditions: None (query). The system shows i and its balance.

CO26 registerPayment(amount, paymentMethod)
Cross references: UC16 Register payment
Preconditions: i is selected (CO25). amount is greater than 0 and no more than the balance.
Postconditions:

- Payment pay was created with pay.amount = amount, pay.date = the current date, and the payment method recorded.
- pay was associated with i (Paid-by), so the balance went down by amount. When the balance reaches 0, the invoice is paid.

CO27 addProfessional(name, professionalInfo, specialtyIds, locationIds, workingHours)
Cross references: UC17 Manage professionals
Preconditions: The given specialties and locations exist.
Postconditions:

- hp was created with hp.name and hp.professionalInfo set.
- hp was associated with the default professional Role (Has).
- Schedule sch was created with sch.workingHours = workingHours and associated with hp (Has).
- hp was associated with each given Specialty (Has) and each given ClinicLocation (Practises-at).

CO28 updateProfessional(professionalId, name, professionalInfo, specialtyIds, locationIds, workingHours)
Cross references: UC17 Manage professionals
Preconditions: hp exists. None of hp's Booked or Confirmed appointments falls outside the new working hours or locations.
Postconditions:

- hp's attributes and Schedule.workingHours took the new values.
- hp's Specialty and ClinicLocation associations were replaced by the given ones.

CO29 assignRole(userId, roleName)
Cross references: UC18 Manage user roles and access
Preconditions: The user (a HealthcareProfessional or AdministrativeStaff) and the Role exist.

Postconditions: The user's association with the old Role was broken, and a new association with the given Role was formed (Has).

CO30 viewActivityLog(fromDate, toDate)

Cross references: UC19 View activity log

Preconditions: None.

Postconditions: None (query). The system returns the Activities between the two dates.